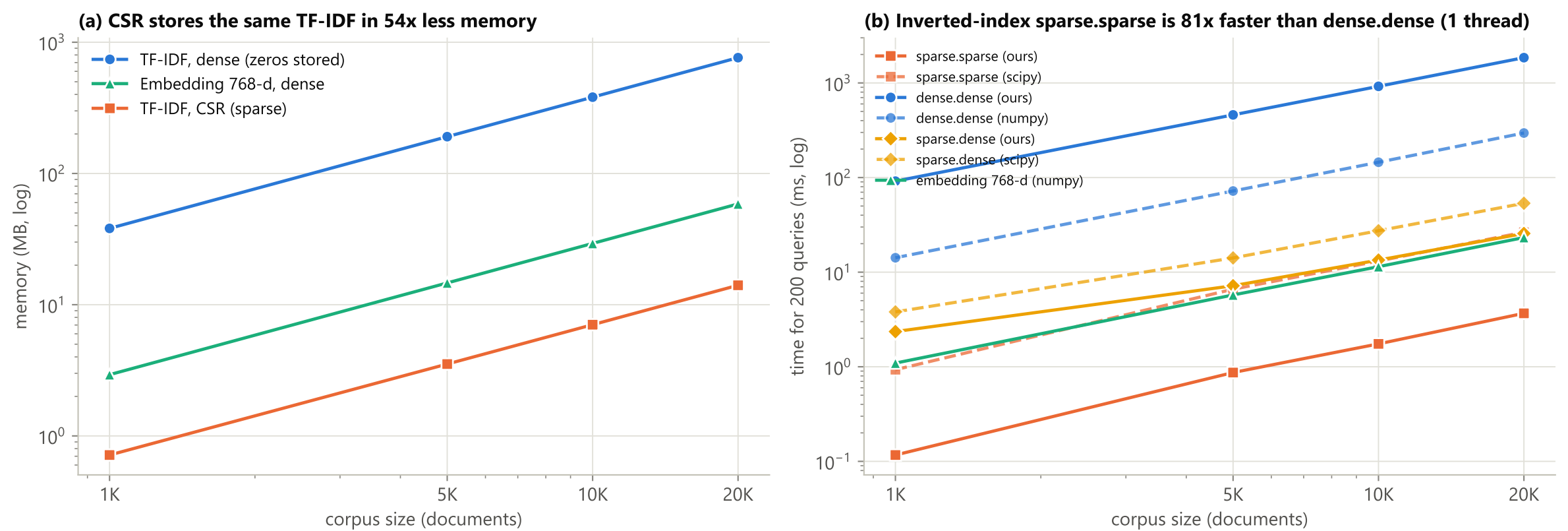
(a) CSR stores the same TF-IDF in 54x less memory
TF-IDF, dense (zeros stored)
Embedding 768-d, dense
TF-IDF, CSR (sparse)
memory (MB, log)
corpus size (documents)
1K
5K
10K
20K
(b) Inverted-index sparse.sparse is 81x faster than dense.dense (1 thread)
sparse.sparse (ours)
sparse.sparse (scipy)
dense.dense (ours)
dense.dense (numpy)
sparse.dense (ours)
sparse.dense (scipy)
embedding 768-d (numpy)
time for 200 queries (ms, log)
corpus size (documents)